PROJECT TITLE

IOT – ENABLED TELEPRESENCE ROBOT FOR REAL- TIME REMOTE PATIENT MONITORING

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Telepresence Robot for Remote Healthcare Monitoring and Emergency Alert System

Abstract

The Telepresence Continuous Monitoring Robot for Remote Healthcare is designed to assist healthcare professionals in monitoring patients remotely by combining robotics, wireless communication, and sensor-based health monitoring technologies. In many hospitals, especially in rural and resource-limited areas, medical staff often face challenges in continuously observing patients due to limited manpower and increasing patient numbers. This project proposes a cost-effective robotic system that can move within hospital environments and allow doctors to observe patients from a remote location. The robot is controlled through a WiFi-based communication system using an ESP8266 microcontroller, which manages the movement of the robot through a motor driver and DC motors. A camera module is integrated into the system to provide live video streaming so that doctors can visually monitor patients and their surroundings in real time without needing to be physically present.

In addition to visual monitoring, the robot is equipped with sensors that measure basic health parameters such as temperature and pulse rate. These readings help doctors assess the patient's condition remotely and identify possible health issues at an early stage. The system also includes an ultrasonic sensor for obstacle detection, which improves safety by preventing collisions while the robot moves within hospital rooms or corridors. An emergency alert mechanism is integrated into the robot, allowing patients to notify healthcare professionals immediately in critical situations by pressing a dedicated button. The RESULT obtained from system testing show that the robot can successfully perform wireless movement, real-time video transmission, health parameter monitoring, obstacle detection, and emergency notification.

The project demonstrates that a practical and affordable telepresence robotic system can be developed using commonly available electronic components and IoT technology. Such systems can help reduce the workload of healthcare staff, improve patient supervision, and increase the efficiency of healthcare services. The developed robot also highlights the potential of telepresence robotics in modern healthcare systems, particularly in areas where access to medical facilities is limited. With further improvements such as autonomous navigation, advanced medical sensors, and improved communication systems, the proposed solution can evolve into a more intelligent and reliable healthcare monitoring platform in the future.

Chapter 1

Introduction

The Telepresence Continuous Monitoring Robot for Remote Healthcare is developed to provide an effective solution for remote patient monitoring, especially in healthcare environments where medical staff and resources are limited. Many hospitals, particularly those located in rural and remote areas, face significant challenges in providing continuous observation for patients due to the shortage of healthcare professionals and the increasing number of patients requiring medical attention. In such situations, doctors and nurses may not always be able to physically check on every patient regularly, which can sometimes delay the identification of health problems. This project introduces a robotic system that can assist healthcare professionals by enabling remote monitoring through advanced yet affordable technology. The robot is designed as a mobile unit that can move inside hospital rooms or wards while being controlled wirelessly using a WiFi-based communication system. By using a microcontroller-based control system, the robot can receive commands from a mobile phone or computer and perform movements such as forward, backward, left, and right navigation. This mobility allows the robot to reach patients in different locations within the healthcare facility. A camera module is integrated into the robot to provide real-time video streaming, allowing doctors to visually monitor patients and their surroundings from a remote location. This feature helps healthcare professionals observe patient conditions without needing to be physically present, thereby saving time and improving efficiency in hospital operations.

In addition to video monitoring, the robot is equipped with important health monitoring sensors that collect basic patient health information. The temperature sensor measures the patient's body temperature, which is an important indicator of various medical conditions such as fever or infection. Similarly, the pulse rate sensor measures the heart rate of the patient, providing valuable information about cardiovascular health and overall physical condition. These health parameters are transmitted through the system so that doctors can monitor them remotely. The integration of these sensors allows the robot to function not only as a telepresence device but also as a basic patient monitoring system. Another important feature of the system is the obstacle detection mechanism implemented using an ultrasonic sensor. This sensor continuously checks for objects in front of the robot and helps prevent collisions by stopping the robot when an obstacle is detected. This ensures safe movement of the robot within hospital environments were equipment, beds, or other objects may be present. The robot also includes an emergency alert system that plays a crucial role in patient safety. In case of a medical emergency or urgent situation, the patient can press an emergency button installed on the robot, which immediately sends a notification or alert to the doctor or healthcare staff. This feature ensures that medical assistance can be provided quickly and efficiently.

Another important aspect of this project is its cost-effective and practical design. Many telepresence robots available in the market are expensive and require advanced infrastructure, which makes them difficult to implement in small hospitals or rural healthcare centres. This project focuses on using affordable and widely available electronic components while still achieving the core functionalities required for healthcare monitoring. The system combines robotics, Internet of Things (IoT) technology, wireless communication, and sensor integration to create a smart healthcare solution that is accessible and scalable. By implementing features such as remote navigation, live video streaming, patient health monitoring, obstacle detection, and emergency alert mechanisms, the proposed robot demonstrates how technology can support medical professionals in improving patient care. The project highlights the potential of telepresence robotics in modern healthcare systems, where remote monitoring, efficient communication, and quick response to emergencies are becoming increasingly important. Overall, the development of this telepresence continuous monitoring robot shows that innovative technological solutions can help bridge the gap between patients and healthcare providers, improve healthcare accessibility, and enhance the quality of medical services in both urban and rural environments.

CHAPTER 2

LITERATURE REVIEW

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This research focuses on the use of telepresence robots to support older adults with dementia. It shows how robotic systems can help improve patient interaction, monitoring, and emotional well-being in healthcare settings.

[3] Kristoffersson, A., Coradeschi, S., & Loutfi, A. (2013).
This paper provides a detailed review of mobile robotic telepresence systems, explaining their design, communication systems, and applications. It is important for understanding how robots can be controlled remotely and used for monitoring purposes.

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This research introduces advanced telepresence robots that use intelligent algorithms to reduce communication delays. It highlights improvements in robot control and performance in remote monitoring systems.

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This paper reviews different telepresence robots used in healthcare, their applications, and challenges. It provides insights into system architecture and how robots can improve remote healthcare services.

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This paper discusses ethical considerations in human–robot interaction such as privacy, trust, and autonomy. These factors are important when implementing robots in healthcare monitoring systems.

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This research describes secure IoT-based healthcare systems that use body sensor networks to monitor patient health data remotely and securely.

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This paper explains how large amounts of healthcare data can be processed using IoT and cloud technologies, which is useful for future improvements in remote monitoring systems.

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This article reviews wireless sensor networks used in healthcare monitoring systems. It explains how sensors can collect and transmit patient health information.

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This paper introduces the Robot Operating System (ROS), a software platform used for building and controlling robots. It is useful for developing advanced robotic applications.

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This research discusses challenges in socially assistive robotics and explains how robots can support people in healthcare and rehabilitation environments.

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This article explains the development of autonomous robotic systems capable of navigating environments independently, which is relevant for future autonomous healthcare robots.

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This paper discusses medical robotics and the ethical, legal, and regulatory considerations involved in developing healthcare robotic systems.

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This research explores wearable health monitoring systems that collect and transmit health data to cloud platforms for analysis and monitoring.

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This paper reviews wearable sensors used in healthcare monitoring systems and explains how they can help track patient health conditions continuously.

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This article reviews methods used to analyze human activities, which can help improve patient monitoring systems using intelligent algorithms.

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This research explains situation awareness in human–robot interaction, which is important for designing robots that can operate effectively in healthcare environments.

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This study provides a survey of socially interactive robots and explains how robots can interact with humans in various environments, including healthcare.

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This book provides comprehensive knowledge about robotics, including robot design, control systems, and applications in various industries including healthcare.

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Chapter 3

Materials and Methodology

This chapter explains the hardware components used in the development of the telepresence healthcare monitoring robot and describes the methodology followed to design, assemble, and implement the system. The objective of this system is to create a mobile robot that can monitor patients remotely, collect health data, and provide live video streaming to healthcare professionals.The system integrates multiple hardware components including a microcontroller, sensors, communication modules, and a robotic movement mechanism. These components work together to enable remote monitoring, obstacle detection, patient health measurement, and emergency alert functionality.

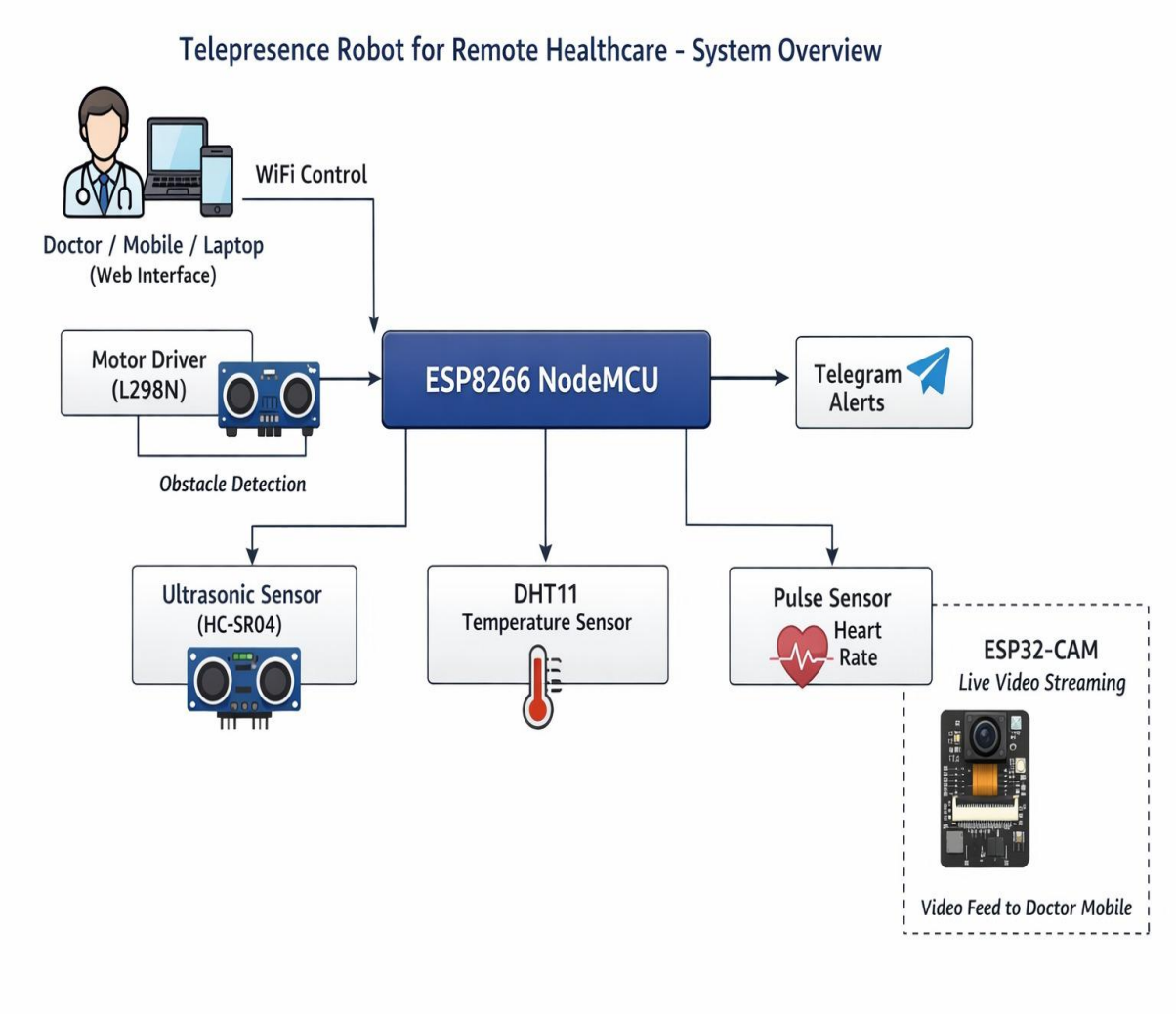


Figure 1: system overview

2.1 Materials Used

The telepresence healthcare monitoring robot is built using the following hardware components:

ESP8266 WiFi Module

The ESP8266 WiFi module acts as the main controller of the robot. It is responsible for processing commands received through wireless communication and controlling the entire system. This module connects the robot to a WiFi network, allowing users to operate the robot remotely using a mobile device or computer.The ESP8266 controls: Motor driver operations Sensor data collection,Communication with external devices,System .coordination.Its built-in WiFi capability makes it suitable for Internet of Things (IoT) applications, including remote healthcare monitoring systems.

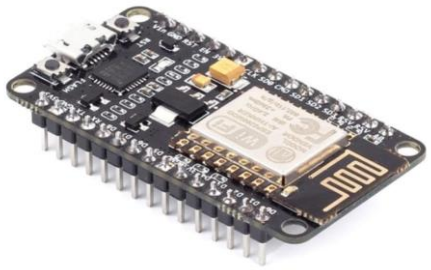


Figure 2 : ESP8266 NodeMCU Wi-Fi Microcontroller Board

L298N Motor Driver

The L298N motor driver module is used to control the movement of the DC motors. It acts as an interface between the ESP8266 microcontroller and the motors. Since the microcontroller cannot directly supply enough current to drive the motors, the motor driver helps amplify the signals and control motor direction and speed.The motor driver enables the robot to perform movements such as: Forward movement,Backward movement,Left turn,Right turn,Stop. This module ensures smooth and controlled navigation of the robot within the hospital environment.



Figure 3 : L298N Dual H-Bridge Motor Driver Module

DC Motors

DC motors are used to drive the wheels of the robot and enable its movement. In this project, two DC motors are connected to the motor driver module. These motors convert electrical energy into mechanical motion, allowing the robot to move in different directions based on commands received from the controller. The motors play a crucial role in enabling the robot to move between patient beds or different hospital locations for monitoring purposes.



Figure 4 : 12V DC Motor for Robot Movement

Chassis and Wheels

The chassis forms the structural framework of the robot and supports all hardware components such as the microcontroller, sensors, battery, and camera module. It ensures that the components are properly mounted and protected during operation. The wheels are attached to the DC motors and allow smooth movement of the robot across the floor surface. A stable chassis and wheel arrangement ensures better balance and mobility of the robot.



Figure 5 : Robot Wheels (Tires) for Movement

ESP32-CAM Camera Module

The ESP32-CAM module is used for capturing live video and streaming it to the doctor or healthcare professional. This module is equipped with a built-in camera and WiFi capability, which allows real-time video transmission. The camera module enables: Visual monitoring of patients, Remote observation by doctors, Improved communication between patient and doctor. This feature is especially useful in situations where direct physical presence of doctors is not possible.

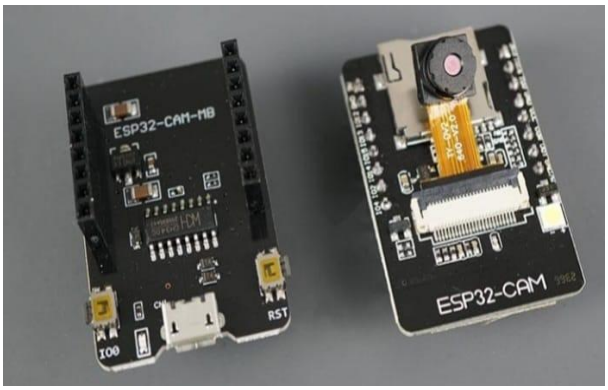


Figure 6 : ESP32-CAM Module with OV2640 Camera and MB Programmer Board

Ultrasonic Sensor (HC-SR04)

The ultrasonic sensor is used for obstacle detection. It works by sending ultrasonic waves and measuring the time taken for the waves to reflect back from an object. Based on this time delay, the distance between the robot and the obstacle is calculated. This sensor helps: Detect objects in front of the robot,Prevent collisions,Improve navigation safety.When an obstacle is detected within a certain distance, the robot automatically stops or changes direction.



Figure 7 : HC-SR04 Ultrasonic Distance Sensor Module

DHT11 Temperature Sensor

The DHT11 sensor is used to measure the temperature near the patient. It provides digital output that can be easily processed by the microcontroller. Monitoring temperature is important in healthcare applications, as it helps detect fever or abnormal body temperature conditions.The temperature readings are displayed in the system and can be monitored remotely by healthcare professionals.

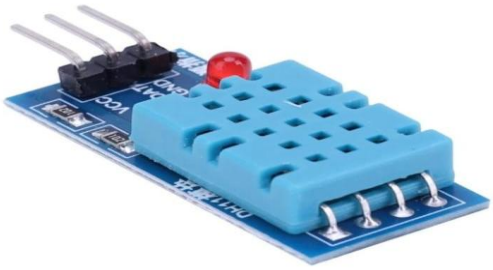


Figure 8 : DHT11 Temperature and Humidity Sensor Module

Pulse Rate Sensor

The pulse rate sensor is used to measure the heart rate of the patient. The patient places their finger on the sensor, and it detects the pulse using light-based sensing technology.The sensor calculates the heart rate in beats per minute (BPM). Monitoring pulse rate helps doctors understand the patient's health condition and detect irregular heart activity.



Figure 10 : Pulse Sensor for Heart Rate Monitoring

Battery Pack

The battery pack provides the necessary power supply to all components of the robot, including motors, sensors, and microcontrollers. A stable power supply is important to ensure continuous operation of the system without interruption.The battery makes the robot portable and allows it to operate without direct electrical connections.



Figure 11 : 7.4V Lithium-Ion Battery Pack (2S Configuration)

Emergency Button

The emergency button is included in the system to allow patients to send an alert signal to doctors in case of critical situations. When the button is pressed, the system sends a notification message through the communication platform used in the project.This feature improves patient safety and ensures quick response during emergencies.



Figure 12 : Two-Pin Push Button Switch

2.2 System Methodology

The development of the telepresence robot was carried out in several stages to ensure proper integration of hardware and software components.

Hardware Assembly

The first stage involved assembling the robot chassis and mounting all hardware components. The DC motors were fixed to the chassis, and the wheels were attached to enable movement. The battery pack and control unit were securely placed on the robot frame.After assembling the physical structure, the motor driver module was connected to the DC motors and the ESP8266 controller.

Microcontroller Programming

The ESP8266 microcontroller was programmed using Arduino IDE. The control logic was written in Embedded C to manage robot movement, sensor readings, and wireless communication.The program includes functions for:Robot movement control,Sensor data reading,Obstacle detection,Communication with mobile devices,Emergency alert system,The code was uploaded to the microcontroller and tested using the serial monitor.

Sensor Integration

In the next stage, the sensors were connected to the microcontroller.The ultrasonic sensor was integrated to detect obstacles and prevent collisions during robot movement. The DHT11 temperature sensor was connected to monitor the patient's temperature. The pulse rate sensor was added to measure heart rate.Each sensor was tested individually to ensure accurate data collection before integrating them into the main system.

Camera Setup and Live Streaming

The ESP32-CAM module was configured to provide live video streaming. The camera was mounted on the robot in a position where it could clearly capture the patient's surroundings.The camera stream was accessed through a web interface, allowing doctors to monitor patients remotely using a smartphone or computer connected to the same network.

Wireless Communication Setup

Wireless communication was established using WiFi technology. The ESP8266 module connects to a WiFi network and receives commands from the user device. These commands control the movement of the robot and enable remote monitoring.This step ensures that the robot can be operated from different locations within the hospital.

Emergency Alert System Implementation

An emergency alert system was implemented using a push button. When the patient presses the button, the system sends an alert message to the doctor through the configured communication platform.This ensures immediate response during medical emergencies.

System Testing

After integrating all components, the complete system was tested. The following tests were conducted:Robot movement test,Obstacle detection test,Temperature monitoring test,Pulse rate monitoring test,Live video streaming test

Emergency alert system test

The system successfully performed all required functions, demonstrating the effectiveness of the telepresence healthcare monitoring robot.

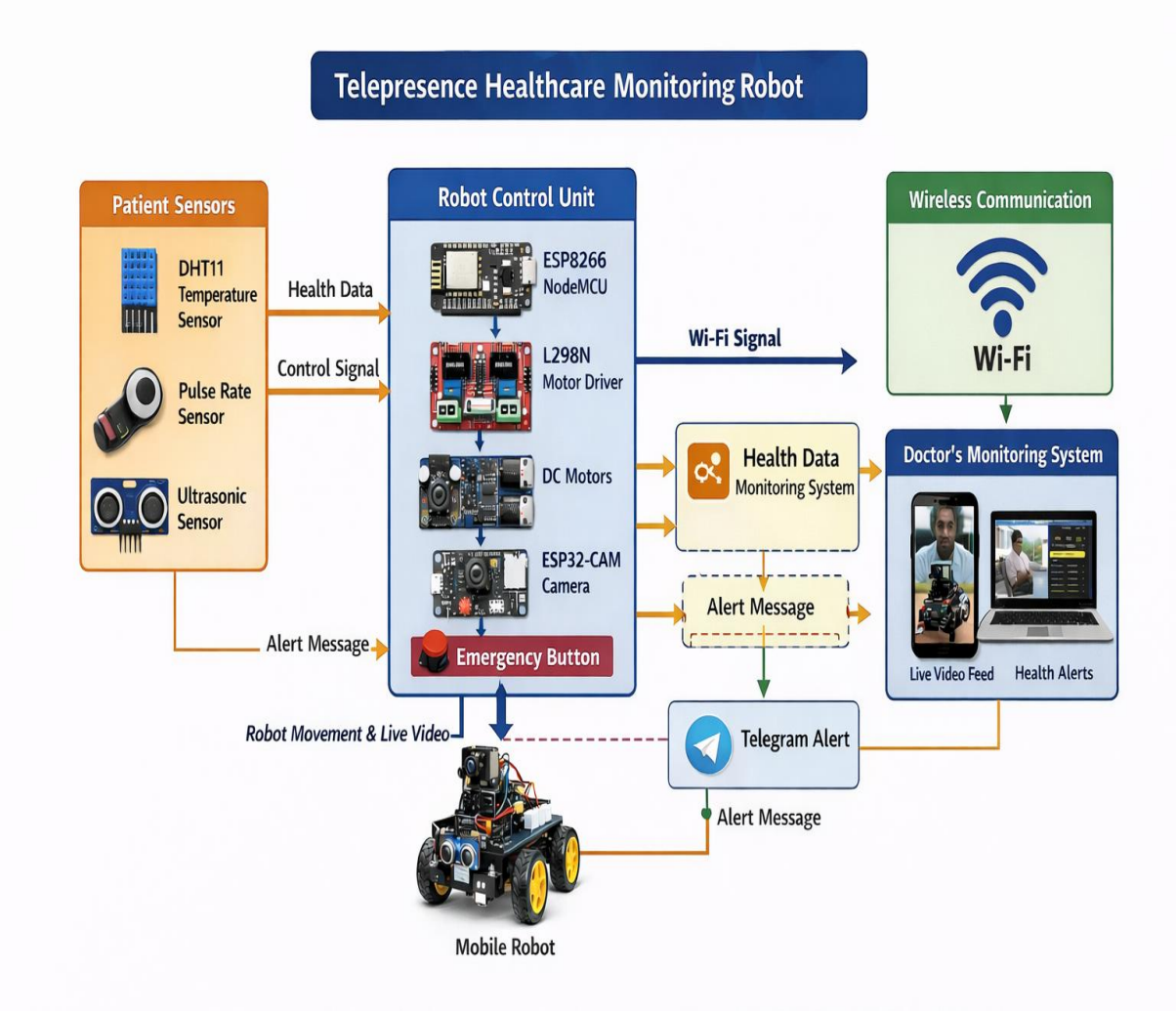


Figure 13: Proposed model

Chapter 4

Result

The Result obtained after the successful development and testing of the Telepresence Continuous Monitoring Robot for Remote Healthcare. The system was tested to evaluate its performance in terms of robot movement, video streaming, sensor data collection, obstacle detection, and emergency alert functionality. Each component of the system was verified to ensure that it operates correctly and supports the overall objective of remote patient monitoring.

3.1 Robot Movement Performance

The movement of the robot was tested using wireless control through the WiFi network. The ESP8266 microcontroller received commands from the user device and transmitted signals to the motor driver module. The motor driver then controlled the DC motors to perform the required movements.During testing, the robot was able to move successfully in all directions, including: Forward movement, Backward movement, Left turn, Right turn, Stop function. The robot responded quickly to control commands and demonstrated stable and smooth navigation. The chassis and wheel arrangement provided proper balance and ensured that the robot moved efficiently on flat surfaces such as hospital floors. The RESULT confirmed that the motor driver and microcontroller integration worked effectively in controlling the robot's motion.

3.2 Live Video Streaming RESULT

The ESP32-CAM module was tested for real-time video streaming functionality. The camera successfully captured live video from the robot’s surroundings and transmitted it to the monitoring device through the wireless network. Doctors or users were able to access the video feed using a web interface on their mobile phones or computers. The video streaming allowed remote visual monitoring of patients and their environment. The video feed was clear under normal lighting conditions and helped demonstrate how telepresence robots can assist doctors in observing patients remotely without being physically present in the same location.

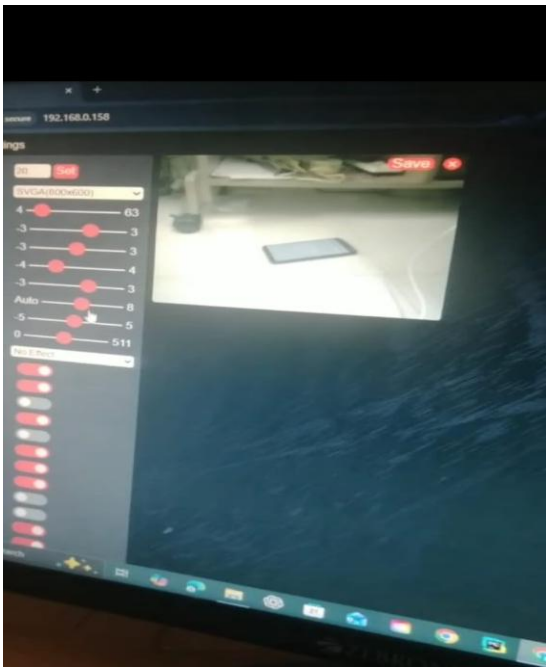


Figure 14: Screenshot-Live video streaming Result

3.3 Temperature Monitoring RESULT

The DHT11 temperature sensor was tested to measure the temperature near the patient. The sensor collected temperature readings and transmitted the data to the ESP8266 microcontroller. The readings were displayed through the serial monitor and system interface. The sensor was able to provide stable and consistent temperature values during testing. This functionality is important for detecting abnormal body temperature conditions such as fever. The successful operation of the temperature sensor confirmed that the system can monitor basic patient health parameters effectively.

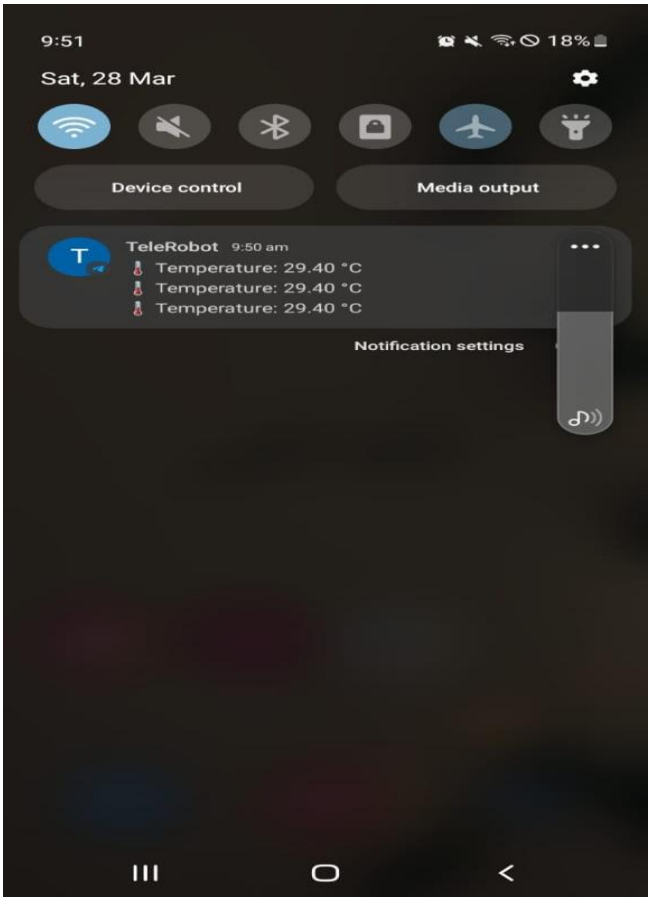


Figure 15: Screenshot- Temperature Monitoring Result

3.4 Pulse Rate Monitoring RESULT

The pulse rate sensor was tested by placing a finger on the sensor surface. The sensor detected the pulse signal and calculated the heart rate in beats per minute (BPM).The system successfully measured the pulse rate and displayed the RESULT. The readings were consistent and demonstrated the capability of the robot to monitor patient heart rate remotely. This feature allows doctors to monitor basic vital signs and understand the patient's health condition without direct physical interaction.

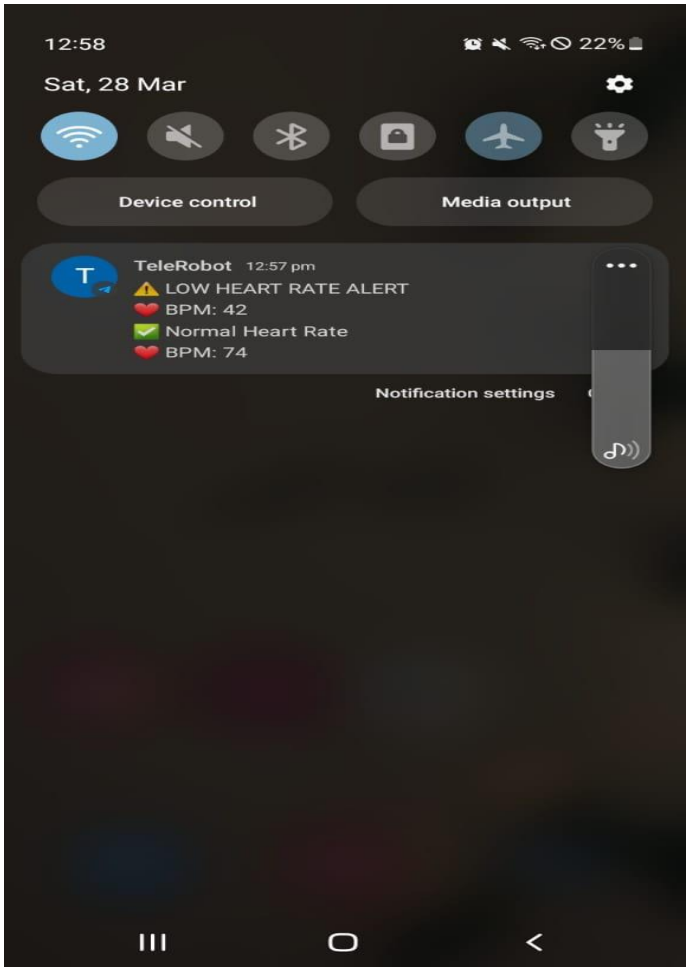


Figure 16: Screenshot- Pulse Rate Monitoring Result

3.5 Obstacle Detection RESULT

The ultrasonic sensor was tested to evaluate its ability to detect obstacles during robot movement. Various objects were placed in front of the robot to simulate real-world obstacles such as walls or equipment. The ultrasonic sensor successfully detected obstacles within a specific distance and transmitted this information to the microcontroller. When an obstacle was detected, the robot automatically stopped moving to avoid collision. This feature improves the safety and reliability of the robot, especially when navigating inside hospital rooms or corridors.

3.6 Emergency Alert System RESULT

The emergency alert system was tested using the emergency push button. When the button was pressed, the system successfully generated an alert notification and sent it to the doctor through the communication platform used in the project. This functionality ensures that patients can quickly notify healthcare professionals during critical situations. The successful testing of the

emergency alert system demonstrated that the robot can assist in providing timely medical attention.

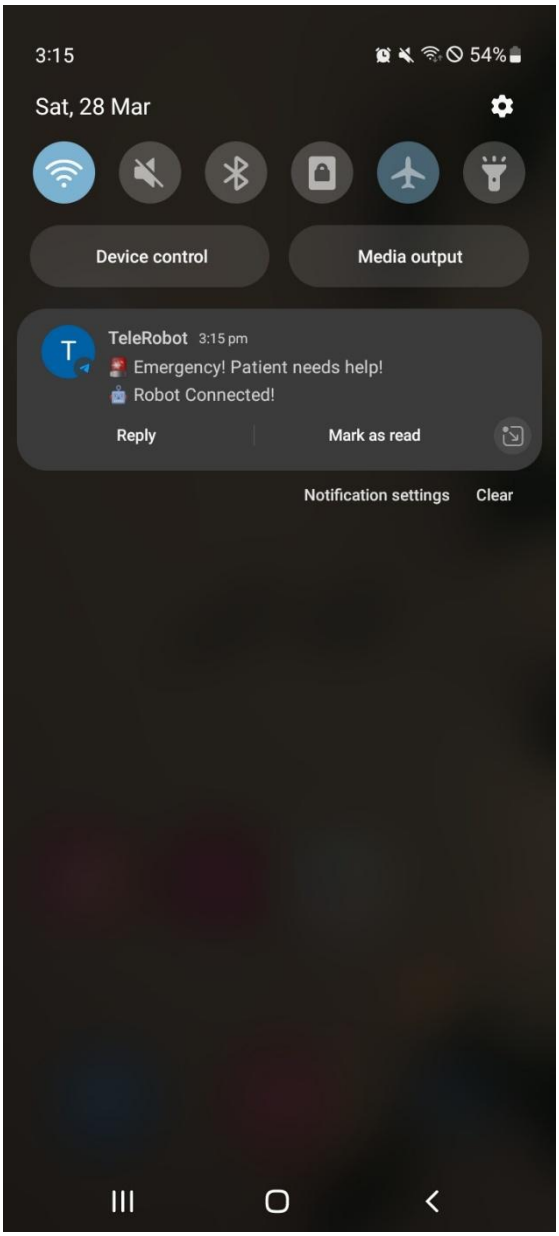


Figure 17: Screenshot- Emergency Alert System Result

3.7 Overall System Performance

After testing all individual components, the entire system was evaluated as a complete telepresence healthcare monitoring robot. The integration of robotics, wireless communication, sensors, and camera modules worked effectively together. The robot was able to: Move smoothly using wireless control, Stream live video in real time**,** Monitor temperature and pulse

rate, Detect obstacles and avoid collisions Send emergency alerts when required The RESULT indicate that the developed system successfully achieved the intended objectives of the project. The telepresence robot demonstrated the ability to assist healthcare professionals in remote patient monitoring and improve healthcare accessibility.

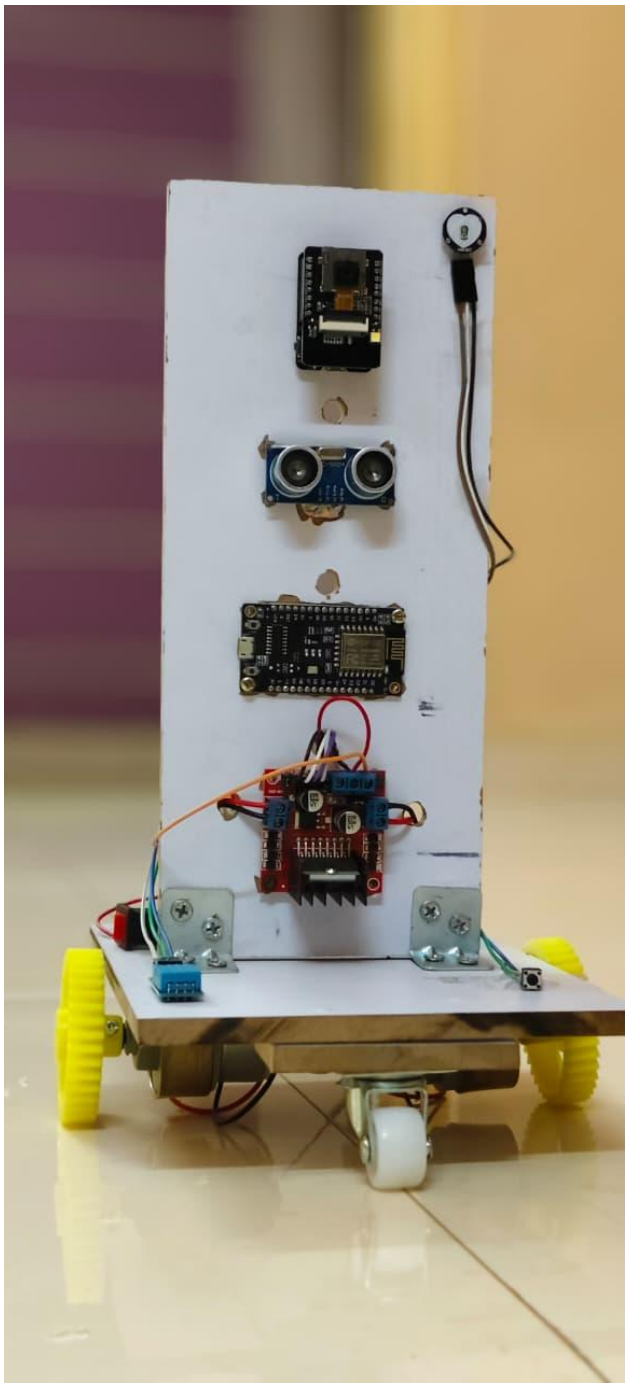


Figure 18: Prototype Model

Chapter 5

Discussion and Conclusion

This chapter discusses the performance of the developed system, analyzes the RESULT obtained during testing, highlights the advantages of the proposed solution, and presents the final conclusion of the project.

5.1 Discussion

The RESULT obtained from the implementation of the Telepresence Continuous Monitoring Robot indicate that the system is capable of supporting remote healthcare monitoring effectively. The integration of robotics, wireless communication, and health monitoring sensors made it possible to develop a system that allows doctors to observe and monitor patients without being physically present in the same location.

During testing, the robot demonstrated stable movement and responded accurately to control commands transmitted through the wireless network. The ESP8266 microcontroller successfully coordinated communication between the motor driver, sensors, and camera module. This shows that the selected hardware components are suitable for implementing an IoT-based healthcare monitoring system.

One of the most important features of the system is live video streaming. The ESP32-CAM module enabled real-time video transmission, which allows healthcare professionals to visually monitor the patient and their surroundings. This feature can be highly beneficial in hospital environments where doctors need to monitor multiple patients at the same time or when direct physical access to patients is limited.

The integration of health monitoring sensors further enhances the usefulness of the system. The DHT11 temperature sensor and pulse rate sensor successfully measured basic health parameters such as temperature and heart rate. These readings provide important information about the patient's condition and can help doctors identify potential health issues at an early stage.

Another important feature implemented in the system is obstacle detection using the ultrasonic sensor. During testing, the robot was able to detect objects placed in its path and automatically stop to avoid collisions. This improves the safety and reliability of the robot, especially when operating in hospital rooms or crowded environments.

The emergency alert system also proved to be effective. When the emergency button was pressed, the system successfully sent a notification to the doctor. This feature is very important in healthcare applications, as it ensures that patients can quickly alert medical staff in case of emergencies.

One of the major advantages of this project is its cost-effectiveness. Many existing telepresence robotic systems available in the market are expensive and require advanced infrastructure. In contrast, the proposed system uses affordable and easily available components, making it

suitable for small hospitals, clinics, and rural healthcare centers. This affordability increases the practicality and scalability of the system.

In addition to affordability, the system is also portable and easy to operate. The robot can move freely within the hospital environment and can be controlled remotely through wireless communication. This reduces the workload on healthcare professionals and allows them to monitor patients more efficiently.

However, despite the successful implementation of the system, some limitations were observed. Currently, the robot requires manual control for navigation. This means that a user must actively control the robot's movement using a mobile device or computer. In future implementations, autonomous navigation could be added to allow the robot to move automatically within hospital environments.

Another limitation is that the system currently monitors only basic health parameters such as temperature and pulse rate. While these parameters are useful, advanced medical monitoring systems often require additional sensors such as blood pressure sensors, oxygen saturation sensors, and ECG monitoring systems. Integrating these sensors would significantly improve the functionality of the robot.

Additionally, the system relies on a stable WiFi connection for communication and video streaming. In environments with poor network connectivity, the performance of the robot may be affected. Therefore, improving communication reliability could be another area of future improvement.

Overall, the system demonstrated good performance and successfully achieved the objectives defined for the project.

5.2 Conclusion

The development of the Telepresence Continuous Monitoring Robot demonstrates how modern technological solutions can be applied to improve healthcare monitoring systems. This project focused on designing a robotic platform that combines mobility, real-time communication, and basic health monitoring capabilities to assist healthcare professionals in supervising patients from a remote location. The successful integration of the microcontroller, camera module, sensors, and wireless communication system shows that a compact and functional healthcare robot can be developed using accessible and cost-effective components.

Throughout the implementation and testing of the system, the robot proved capable of performing the essential tasks required for remote patient observation. The ability to transmit live video enables doctors to visually assess patient conditions without being physically present, while the integration of health monitoring sensors allows the system to collect important physiological data. The inclusion of obstacle detection ensures safe navigation within indoor environments, and the emergency alert mechanism adds an additional layer of patient safety by enabling quick communication during urgent situations. These capabilities indicate that the robot can function as a supportive tool in healthcare environments where continuous monitoring is required.

Another important outcome of this project is the demonstration that affordable robotic solutions can be developed for healthcare applications. Instead of relying on complex and high-cost medical robotic systems, this project shows that essential monitoring functions can be achieved using simple hardware architecture and efficient system design. This approach makes the system more suitable for small hospitals, clinics, and healthcare centers that may not have access to advanced medical technologies.

Overall, the project confirms that telepresence-based robotic systems can contribute to improving patient supervision, supporting healthcare professionals, and enhancing the efficiency of healthcare services. The implementation of this system provides a foundation for future improvements and encourages further research in the development of intelligent healthcare robots. With continuous advancements in communication technologies and sensor systems, telepresence robots are expected to become an important component of modern healthcare infrastructure in the coming

Chapter 6

Future Work

The Telepresence Continuous Monitoring Robot developed in this project demonstrates the basic functionality required for remote healthcare monitoring. However, technology in the fields of robotics, IoT, and healthcare systems is continuously evolving, and there are several improvements that can be made in the future to enhance the performance and capabilities of the system. Future developments can focus on improving the intelligence of the robot, expanding health monitoring features, and increasing the reliability of communication systems.

One of the major areas for future improvement is the implementation of autonomous navigation. In the current system, the robot movement is controlled manually through wireless commands. In future versions, the robot can be designed to move automatically within hospital environments using intelligent navigation systems. By integrating mapping techniques and advanced sensors, the robot could identify hospital rooms, corridors, and obstacles on its own. This would allow the robot to reach patients automatically and reduce the need for constant human control.

Another important improvement that can be made in the future is the addition of more advanced medical monitoring sensors. At present, the system measures only temperature and pulse rate. However, modern healthcare monitoring systems require more detailed patient data. Future versions of the robot can include sensors to measure blood pressure, oxygen levels in the blood, respiration rate, and electrocardiogram signals. These additional health monitoring capabilities would allow doctors to better understand the patient's health condition and provide more accurate medical support.

Future development can also focus on enhancing the video monitoring system. The quality of video streaming can be improved by using higher resolution cameras and more efficient video transmission technologies. Features such as night vision cameras or automatic camera adjustment could also be implemented so that monitoring can continue effectively under different lighting conditions.

Another possible improvement is the integration of cloud-based data storage and monitoring systems. Patient health data collected by the robot can be stored securely in cloud platforms, allowing doctors to access patient records from anywhere. This would also help in maintaining historical health data, which can be useful for long-term patient monitoring and medical analysis.

The system can also be enhanced by developing a mobile application with a more advanced user interface. A dedicated mobile application could allow doctors to control the robot more easily, view live video streams, and receive alerts or notifications instantly. Additional features such as voice commands or automated alerts could further improve usability.

Improving the communication reliability of the system is another important future enhancement. The current system depends mainly on WiFi networks, which may sometimes experience signal interruptions. In the future, the robot can be integrated with more stable

communication technologies such as hybrid network systems or advanced IoT communication protocols to ensure continuous connectivity.

Another potential development is the use of intelligent monitoring and data analysis techniques. Future systems could analyze patient health data automatically and detect abnormal conditions. For example, if the patient's heart rate or temperature exceeds a safe limit, the system could automatically notify doctors or healthcare staff. This would improve response time during emergencies and enhance patient safety.

The physical design of the robot can also be improved in the future. The robot can be made more compact, lightweight, and efficient so that it can operate easily in crowded hospital environments. In addition, improving battery capacity would allow the robot to function for longer periods without frequent charging.

Furthermore, future research can explore the possibility of integrating this system with smart hospital infrastructure, where multiple robots operate together within the same healthcare environment. Such integration could support better hospital management and enable doctors to monitor several patients simultaneously.

Overall, the future improvements mentioned above can significantly enhance the capabilities of the telepresence healthcare robot. With advancements in robotics and healthcare technology, the system can be further developed into a more intelligent and comprehensive healthcare monitoring solution.

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